Data Scientist & Economic Analyst

Rodrigo França Chaves

Chicago, IL, USA · rodrigofrancachaves99@gmail.com · rchaves@uchicago.edu · +1 312 217 5526

Website · Google Scholar · LinkedIn · GitHub · Brazilian Citizen — F1 Visa (STEM OPT Eligible)

Applied economist and data scientist working on cities: property taxation, housing markets, transit, utility regulation, and crime. First author of a peer-reviewed article in *Utilities Policy* on the health effects of water privatization across two decades of Brazilian municipal data. Builds evidence from sources that resist analysis — assessment rolls, transaction records, disease registries, century-old archives, satellite imagery — pairing quasi-experimental identification with machine learning at scale, and explaining the result to people who did not build the model.

EDUCATION

M.S. in Computational Analysis and Public Policy — *Sep 2025 – Jun 2027*

The University of Chicago, Chicago, IL

Specializations: Markets & Regulation; Health Policy

Coursework: Machine Learning for Public Policy, Applied Econometrics, Data Engineering, Program Evaluation, Public Finance, Statistical Computing

Postgraduate Degree in Data Science (Pós-Graduação) — *Sep 2022 – Apr 2024*

Insper – Institute of Research and Education, São Paulo, Brazil

Coursework: Statistical Inference, Econometrics, Machine Learning, Cloud Computing, Predictive Modeling

B.A. in International Relations — *Feb 2018 – Jun 2022*

Ibmec – Brazilian Institute of Capital Markets, Belo Horizonte, Brazil

Coursework: Political Economy, International Trade, Macroeconomics, Quantitative Methods, Public Policy, Political Science

PUBLICATIONS & WORKING PAPERS

Private Ownership of Water and Wastewater Systems: Assessing Health Impacts

Utilities Policy — Peer-Reviewed Journal Article (first author)

Evaluates whether Brazil's 2020 opening of water and sanitation to private operators improved public health, applying Callaway–Sant'Anna staggered difference-in-differences to propensity-matched municipalities over 1998–2021 (322 privatizations recorded nationally). Finds significant reductions in diarrheal morbidity among children under five (ATT -11.6 for infants, -15.8 at ages 1–5) but no effect on non-diarrheal disease; results are mixed across municipalities, with the most robust evidence from Tocantins' transfer to full private ownership. First of two authors; CRediT credits me with the original draft, software, and data curation.

Resistance to Doing Right: Rising Cost of Misconduct from Body-Worn Camera Adoption

Working Paper — In Progress (second author)

Estimates how body-worn cameras changed police behavior in São Paulo, using confidential Military Police microdata (2020–2023) and the staggered rollout of 10,125 cameras across 64 battalions. Battalion boundaries are never publicly released, so I reconstructed them by clustering geocoded incidents into Voronoi polygons — the step that made spatial treatment assignment possible. Callaway–Sant'Anna DiD finds confrontations fell (ATT -0.50, 1%), concentrated in low-income areas (5%; no significant effect in high-income), while property crime rose, most robustly for vehicle theft: de-policing rather than a shift toward procedurally correct enforcement.

WORK EXPERIENCE

Data Science & Research Intern — DECDI (Formerly DIME) — *Summer 2026*

World Bank, Washington, DC

- Reconstructed four decades of Mumbai property tax and land-use policy into one consistent timeline, then tested millions of assessment records for bunching at policy thresholds to establish whether declared values showed statistical signatures of manipulation.
- Quantified how Mumbai's market for additional floor area splits across regulatory instruments (TDR, Premium FSI, Fungible FSI) over two decades of transaction records, identifying which mechanism developers actually use and how market, regulatory, and political conditions drive that choice.
- Measured the built-environment response to Dakar's Bus Rapid Transit corridor using difference-in-differences on a 100 m grid built from Google Open Buildings 2.5D Temporal and GHSL rasters. Because the line was announced in 2018 and opened in 2024, the design identifies the anticipation and construction-phase response, not the effect of operation.
- Found and corrected a building-detection threshold set at nearly double the value the source paper optimizes for Africa. The old cut discarded genuine building stock hardest in dense and informal areas — a neighborhood-varying bias that does not difference out of a DiD. Shipped the pipeline with 200 tests and an interactive explorer for the team.

(Findings confidential to the World Bank.)

Graduate Research Assistant *Mar – Jun 2026 · Resuming Sep 2026*
Mansueto Institute for Urban Innovation, UChicago, Chicago, IL

- Built an end-to-end digitization pipeline — page segmentation, OCR, structured extraction — for century-old cloth-bound fire insurance registers and Library of Congress holdings, converting records that had never been machine-readable into analysis-ready data.
- Extracted thousands of firm addresses from hundreds of register pages and linked them to three historical censuses by street, then applied NLP to firm-name patterns to classify establishments across hundreds of industries — producing a dataset of industrial location and organization in 20th-century Chicago that did not previously exist.
- Returning in September 2026 to write the method up as a research paper documenting the pipeline for replication by other archival-data researchers.

Research Assistant *Aug 2023 – Jul 2025*
Insper Cities Lab (Arq.Futuro), São Paulo, Brazil

- Two years of quasi-experimental policy evaluation across utilities, crime, property taxation, regulation, and real estate markets, working with municipal panels running to millions of records on home prices, tax assessments, and disease incidence.
- First-authored the resulting *Utilities Policy* article on water privatization and public health, and co-authored an in-progress working paper on police body cameras.
- Reconciled administrative microdata from multiple government sources, resolving inconsistent identifiers and coverage gaps to make municipal panels comparable across two decades — the step that made the privatization analysis identifiable at all.
- Mentored undergraduate researchers on research design, econometric methodology, and academic writing; coordinated cross-functional teams under firm publication deadlines.

Teaching Assistant — GIS, Econometrics & R *Aug 2023 – Jul 2025*
Insper, São Paulo, Brazil

- Taught GIS and R to 50+ students, applying spatial analysis to São Paulo's real estate market, transit infrastructure, population density, and urban models of settlement — density and land-value gradients from the center outward.
- Advised undergraduate theses through Brazil's required capstone; one placed second in its cohort.
- Supervised an intensive field course requiring direct coordination with municipal officials.

PROJECTS

NYC Property Tax: A Horizontal-Equity Audit `Python · LightGBM · scikit-learn`
Horizontal-equity audit of New York's assessment roll, testing whether comparable properties carry comparable tax burdens. Benchmarking all 1.1M properties against their peer-group median assessed value per square foot found roughly 43% sitting more than 15% away from their peers — 245k above, 227k below. Within one peer group of 1940s Queens two-family homes worth about $750k, assessments ranged from $12.09 to $45.88 per square foot: a near-fourfold gap in effective tax rate produced by purchase date under NYC's Class 1 assessment cap rather than by any difference in the properties. Tightening peer groups from quintile to ventile value bins cut within-group price spread from $475k to $48k, trading a small F1 loss for labels that carry meaning — the judgment call the project turned on. Six peer dimensions with a five-level fallback, 138 leak-free engineered features, LightGBM, and a Streamlit prototype for audit triage. On a three-person team I owned the labeling methodology, the data pipeline, and model iteration. [GitHub] [Live Dashboard]

Data Centers Next Door: Cloud Infrastructure and Housing Costs `R · Shiny · Spatial Econometrics`
An exploratory tool for examining how data center proximity relates to Chicago housing and household costs (2000–2025), linking scraped facility records and permit-derived opening dates to ZIP-level Zillow prices and ACS utility-cost data. Reports a composite pre/post index by decile rank and z-score. Descriptive by design: it surfaces and maps the association for users to interrogate rather than asserting a causal effect. On a four-person course project I scraped and geocoded the facility data and built the Shiny dashboard. [GitHub] [Live Dashboard]

TECHNICAL SKILLS

Methods: Causal Inference (DiD, Staggered DiD, RDD, IV), Panel & Spatial Econometrics, Program Evaluation, Predictive Modeling, Gradient Boosting, Remote Sensing

Programming & Data: Python, R, SQL, PostgreSQL, Stata; large-scale data pipelines, OCR & archival extraction, geospatial workflows

Tools: QGIS, ArcGIS, Git, LaTeX, Excel, Streamlit, Shiny

Domain: Property tax & valuation, zoning & land use, housing & infrastructure policy, utility regulation, regulatory & compliance analysis

Languages: Portuguese (Native), English (Fluent — TOEFL 112/120), Spanish (Conversational), German (Basic)